

03

/ 13 LECTURES

IP Addresses and Subnetting — Part I

Why IP addresses exist, binary $\leftrightarrow$ decimal fluency, network vs host ID, subnet masks, the AND trick, CIDR notation, public vs private ranges, and the loopback address.

Why IP

IPv4 Format

Binary Refresher

Network ID

Host ID

Subnet Mask

AND Operation

CIDR Notation

Public vs Private

Loopback

APIPA

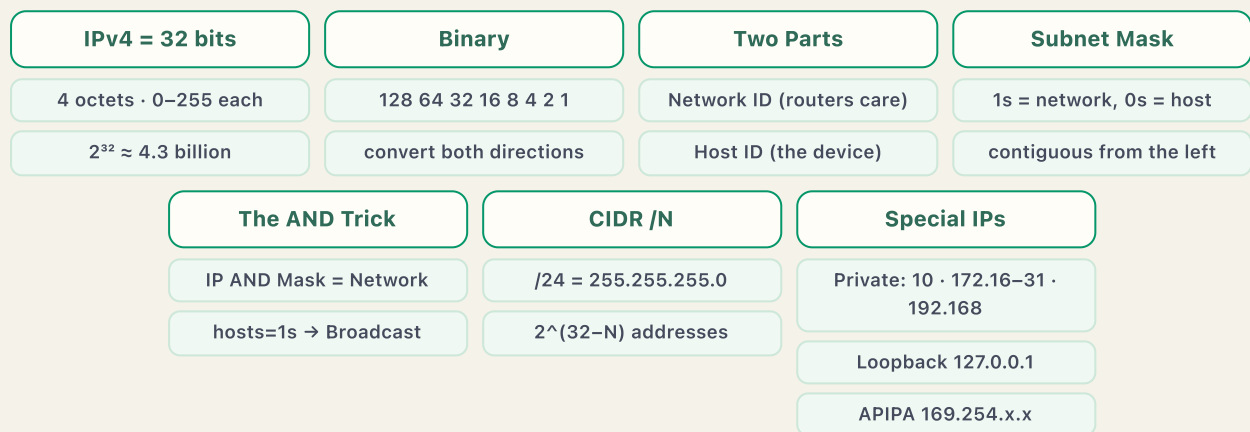
WHAT THIS LECTURE GIVES YOU

The single most exam-generating topic in the course: reading an IPv4 address like a router does. You'll convert binary fluently, split any address into network and host parts with a subnet mask, compute network/broadcast addresses with the AND operation, read CIDR /N notation, and memorise the private ranges — with three fully worked examples.

LEARNING OBJECTIVES — AFTER THIS LECTURE YOU CAN

- Explain why we need IP addresses on top of MAC addresses
- Read and write IPv4 addresses in **binary and dotted-decimal**
- Apply **subnet masks** and **CIDR /N** notation
- Identify **network address, broadcast address, first/last host** of any subnet
- Classify addresses as **public, private, or loopback** — and spot APIPA

IP Addressing I



1 Why IP Addresses?

MAC addresses (Lecture 2) only work **locally** — within one network segment. To route data across the Internet, across thousands of networks, we need a global addressing system. IP addresses provide:

Global uniqueness

Every public IP is unique world-wide — assigned by registries (IANA → regional registries → ISPs).

Hierarchy

The network portion enables routers to aggregate millions of destinations into few routing entries.

Logical & flexible

Reassignable in software — unlike the hardware-burned MAC.

ANALOGY **MAC = your name** (travels with you, meaningless for postal delivery). **IP = your postal address** (hierarchical: city → street → house — exactly how routers narrow the search). A person can move houses (new IP) keeping the same name (MAC).

2 IPv4 Format



■ Four octets (0–255 each) = 32 bits total ■ Total address space $2^{32} = 4,294,967,296 \approx 4.3$ billion

Figure 1. Dotted-decimal notation — four 8-bit octets separated by dots.

3 Binary ↔ Decimal — the Essential Skill

128 bit 7 64 bit 6 32 bit 5 16 bit 4 8 bit 3 4 bit 2 2 bit 1 1 bit 0

Figure 2. Memorise these 8 position values — every conversion is just adding subsets of them.

Decimal → Binary (subtract from the left)

192: $128 \leq 192 \rightarrow \text{bit}=1$, rem $64 \rightarrow 64 \leq 64 \rightarrow \text{bit}=1$, rem 0 $\rightarrow$ rest 0
 $192 = 11000000$

168: $128 \rightarrow 1$ (rem 40) · $64 \rightarrow 0$ · $32 \rightarrow 1$ (rem 8) · $16 \rightarrow 0$ · $8 \rightarrow 1$ (rem 0)
 $168 = 10101000$

Binary → Decimal (add the 1-positions)

11000000 = $128 + 64 = 192$

10101000 = $128 + 32 + 8 = 168$

Full IP: $192.168.1.5 =$

11000000.10101000.00000001.00000101

Decimal	Binary	Decimal	Binary	Decimal	Binary
0	00000000	10	00001010	192	11000000
1	00000001	64	01000000	255	11111111
2	00000010	128	10000000	172	10101100

4 Network ID and Host ID

Every IP address has two parts, split by the subnet mask:

NETWORK PORTION identifies which network — routers route on this

HOST the device

Figure 3. $192.168.1.5 \rightarrow$ network $192.168.1$ · host $.5$ (with a /24 mask).

Historical classes (know the ranges, CIDR replaced them)

Class	First octet	Network bits	Host bits	Default mask	Typical use
A	1–126	8	24	255.0.0.0	giant networks / ISPs
B	128–191	16	16	255.255.0.0	large organisations
C	192–223	24	8	255.255.255.0	small networks

NOTE 127.x.x.x is reserved (loopback) so class A "ends" at 126 in practice. Classful addressing is legacy, but the ranges still appear in exams and tools.

5 Subnet Mask & the AND Operation

A **subnet mask** is a 32-bit number: **1s mark network bits, 0s mark host bits** — and the 1s are always **contiguous from the left**.

11111111.11111111.11111111.00000000

```
/26 = 26 ones = 111...1000000
Block: 192.168.1.64 → hosts .65-.126
Broadcast .127 · usable = 26-2 = 62
```

CIDR	Subnet mask	Usable hosts	Common use	CIDR	Subnet mask	Usable hosts	Common use
/8	255.0.0.0	16,777,214	Large ISPs	/26	...255.192	62	Small subnets
/16	255.255.0.0	65,534	Enterprises	/27	...255.224	30	Small offices
/24	255.255.255.0	254	Home / office	/28	...255.240	14	Very small
/25	...255.128	126	Medium subnets	/30	...255.252	2	Point-to-point

EXAM Memorise the "jump" pattern in the last octet: /24→.0, /25→.128, /26→.192, /27→.224, /28→.240, /29→.248, /30→.252. Each extra /1 bit halves the block: 256 → 128 → 64 → 32 → 16.

7 Public vs Private Addresses

Private range	CIDR	Size	Typical use
10.0.0.0 – 10.255.255.255	10.0.0.0/8	16.7 M	Large corps, cloud VPCs
172.16.0.0 – 172.31.255.255	172.16.0.0/12	1.04 M	Medium networks
192.168.0.0 – 192.168.255.255	192.168.0.0/16	65,536	Home routers, offices

- **Private = not routable on the Internet** — freely reusable inside any organisation; millions of homes all use 192.168.x.x.
- **Public = everything else** — globally routable, assigned by ISPs/registries. Only your router's WAN side holds one (NAT, Lecture 11).

TRAP **172.20.5.1 is private** (172.16–31 range) but **172.32.0.1 is public!** The /12 range stops at 172.31.255.255. Examiners love this.

8 Special Addresses

Address / range	Name	What it means
127.0.0.1 (127.0.0.0/8)	Loopback / localhost	Packets loop back to the same device — never touch the network. ping 127.0.0.1 tests your TCP/IP stack. http://localhost:3000 for local dev servers.
0.0.0.0	Unspecified / "any"	All interfaces (servers bind here) or "no address yet" (DHCP discover)
169.254.0.0/16	APIPA	Self-assigned when DHCP fails — seeing this = "my device couldn't get an IP from the router"
255.255.255.255	Limited broadcast	All hosts on the local subnet; routers never forward it

9 Worked Examples (Exam Pattern)

Example 1 — analyse 192.168.10.130/25

```
/25 mask = 11111111.11111111.11111111.10000000 = 255.255.255.128
Last octet: 130 = 10000010 → first bit (network) = 1 → belongs to upper half
Network: 192.168.10.128      Broadcast: 192.168.10.255
Hosts: 192.168.10.129 – 192.168.10.254      Usable:  $2^7 - 2 = 126$ 
```

Example 2 — is 172.20.5.1 private?

```
172.20.x.x falls in 172.16.0.0 – 172.31.255.255 → YES, private (172.16.0.0/12)
```

Example 3 — network address of 10.45.60.100/20

```
/20 mask = 255.255.240.0 → third octet has 4 network bits + 4 host bits
60 = 00111100 → keep first 4 bits (0011), zero the rest → 00110000 = 48
Network: 10.45.48.0      Broadcast: 10.45.63.255      Usable:  $2^{12} - 2 = 4094$ 
```

SPEED TIP For /20 in the third octet: block size = $256 - 240 = 16$ → network octets are 0,16,32,**48**,64... 60 sits in 48–63 → network .48, broadcast .63. This "block-size jump" trick solves any subnet question in seconds.

LECTURE 3 — RAPID REVISION CARD

IPv4 = 32 bits, 4 octets 0–255; space = $2^{32} \approx 4.3$ B.

Binary positions: 128 64 32 16 8 4 2 1 — add/subtract to convert.

Subnet mask: 1s = network, 0s = host, contiguous left.

Network = IP AND mask; broadcast = host bits all 1; usable = $2^{(\text{host bits})} - 2$.

CIDR /N: N network bits; /24→.0, /25→.128, /26→.192, /27→.224, /28→.240, /30→.252.

Classes: A 1–126 (/8) · B 128–191 (/16) · C 192–223 (/24).

Private: 10/8 · 172.16–31/12 · 192.168/16 — never Internet-routable.

Loopback 127.0.0.1 = self · **169.254.x.x** = DHCP failed · **0.0.0.0** = any.

Important Questions — Exam Practice

A. Multiple Choice

Q1. Which of these is a **private** IP address?

MCQ

- a) 172.32.10.1
- b) 192.169.1.5
- c) 10.200.3.4
- d) 8.8.8.8

Q2. How many usable host addresses exist in a /28 subnet?

MCQ

- a) 16
- b) 14
- c) 30
- d) 254

Q3. The network address of 192.168.1.77/26 is:

MCQ

- a) 192.168.1.0
- b) 192.168.1.64
- c) 192.168.1.32
- d) 192.168.1.63

Q4. A device showing 169.254.10.5 most likely means:

MCQ

- a) It is a public web server
- b) DHCP failed
- c) It is a loopback test
- d) It is a router

Q5. 255.255.255.240 as CIDR is:

MCQ

- a) /26
- b) /27
- c) /28
- d) /29

B. Short Answer (2–4 marks)

Q6. Convert 172 to binary. Show working.

SHORT

Q7. What is the network address of 10.20.30.40/16? Show the AND operation.

SHORT

Q8. Why can't the network address or broadcast address be assigned to a device?

SHORT

Q9. Distinguish public vs private IPs. Why do private ranges exist at all?

SHORT

Q10. What is the loopback address and give two practical uses.

SHORT

C. Long Answer / Numericals (5–8 marks)

Q11. For 172.16.90.77/20 compute: subnet mask, network address, broadcast address, first & last usable host, total usable count. Show the binary of the relevant octets.

NUMERICAL

Q12. Explain the structure of an IPv4 address with a diagram: octets, network portion, host portion, and how the subnet mask divides them. Why is the AND operation the natural way to find the network?

LONG

Q13. You are given 192.168.1.5/24. Write the complete 32-bit binary of the address AND of its subnet mask, then compute the network and broadcast addresses. **NUMERICAL**

Q14. Your friend says "172.25.10.5 can't be private because it doesn't start with 10 or 192.168." **SHORT**
Correct them precisely.

ANSWER KEY

A1. c — 10.x.x.x is private; 172.32 is outside 172.16–31; 192.169 is outside 192.168.

A2. b — $2^4 - 2 = 14$.

A3. b — /26 blocks: 0, 64, 128, 192 → 77 sits in 64–127.

A4. b — APIPA self-assignment when DHCP fails.

A5. c — $240 = 11110000 \rightarrow 28$ ones.

A6. $172 = 128 + 32 + 8 + 4 = 10101100$.

A7. Mask /16 keeps first two octets → **10.20.0.0**.

A8. Network = the subnet's own identity (routers use it); broadcast = "all hosts here". Both are conventions every host must agree on — assigning them breaks addressing.

A9. Public = globally routable, registry-assigned, unique. Private = reusable internally, not routed on Internet. They exist because 4.3B addresses < devices on Earth — reuse + NAT (L11) stretches IPv4.

A10. 127.0.0.1 — packets never leave the machine. Uses: test the TCP/IP stack; run local dev servers (localhost:3000).

A11. /20 → mask 255.255.240.0; third octet 90 = 01011010 → keep 0101 → 01010000 = 80. Network **172.16.80.0**, broadcast **172.16.95.255**, hosts .80.1–.95.254, usable **4094**.

A12. Use Figures 1 & 3; AND zeroes host bits because "anything AND 0 = 0" — leaving exactly the network identity all subnet devices share.

A13. 11000000.10101000.00000001.00000101 AND 11111111.11111111.11111111.00000000 → network **192.168.1.0**, broadcast **192.168.1.255**.

A14. The /12 private block is 172.16.0.0–172.31.255.255 — 172.25.x.x lies inside it, so it IS private. Private ≠ only 10/8 & 192.168/16.

Wrap-Up — carry these into Lecture 4

ANATOMY 32 bits = 4 octets; the mask draws the line between **network bits** (shared identity) and **host bits** (who you are inside it).

AND TO FIND HOME IP **AND** mask = network address. "Anything AND 0 = 0" zeroes host bits, leaving exactly the subnet's identity.

SPECIAL ADDRESSES All-host-bits-0 = network, all-ones = broadcast — never assign them. 127.0.0.1 talks to yourself.

PRIVATE BLOCKS 10/8, 172.16/12, 192.168/16 are reusable inside networks; public addresses are globally unique. NAT (L11) bridges the two.

NEXT **Lecture 4 — IP Addresses & Subnetting II:** borrowing host bits to carve subnets, VLSM, and the speed method that solves any paper question in 30 seconds.